Attack-Free Early Warning and Timely PGD Escalation for Catastrophic Overfitting

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Abstract*— Single-step adversarial training is efficient but can abruptly lose robustness through catastrophic overfitting (CO). We study an online safety trigger based on the cosine consistency of two input gradients already produced by an FGSM parameter update. The CIFAR-10 threshold is selected on seeds 17, 23, and 42 and frozen before held-out evaluation. It detects 7/7 held-out CIFAR-10/PreActResNet-18 events within 60 updates. With the threshold unchanged, it detects 4/5 CIFAR-10/ResNet-18 events, including 2/5 within 60 updates. After one dataset-specific calibration, it detects both observed held-out CIFAR-100 events; the three CIFAR-100 non-event trajectories contain one alarm episode. Eight randomized-FGSM runs at $\epsilon = 8/255$ remain stable and produce no alarms. At $\epsilon = 16/255$, all three stress-test runs naturally collapse; the frozen CIFAR-10 detector warns 20–60 updates beforehand, and the tested epoch-replay response raises terminal PGD-10 accuracy from 0 to $26.48 \pm 0.83\%$. A five-seed timing audit finds that immediate switching, with or without epoch replay, starts PGD-2 before CO in 5/5 cases, versus 3/5 for next-epoch escalation. Without replay, final PGD-10 accuracy is $47.05 \pm 0.74\%$; its $+0.22$-point difference from replay has a paired 95% interval crossing zero. Immediate action provides the timing benefit, while rollback adds no reliable endpoint gain. In controlled reimplementations, GradAlign achieves higher robustness, while the proposed policy uses lower measured core training time. For the fixed seed-101 no-replay checkpoint, full-test PGD-50, five-restart PGD-20, and fixed-subset AutoAttack-standard accuracy are 45.52%, 45.66%, and 42.30%, respectively. Gradient consistency is used as an empirical warning signal; conditional PGD-2 supplies the recovery.

I. Introduction

Deep classifiers can change their predictions under small adversarial perturbations [1], [2]. Adversarial training addresses this failure by optimizing against perturbed examples and remains a strong empirical defense [3]. Its main obstacle is cost because multi-step training attacks require several forward-backward passes for nearly every parameter update.

Fast adversarial training replaces the inner loop with a single FGSM step. With randomized initialization and a suitable schedule, it can approach the robustness of PGD training at much lower cost [4]. Yet single-step training can exhibit *catastrophic overfitting* (CO): iterative-attack accuracy falls sharply toward zero while clean or FGSM accuracy remains high. FastAdv+ responds using periodic PGD validation [5]; GradAlign regularizes local input-gradient geometry at every update [6]. These approaches improve reliability but add attack or gradient computation.

We ask whether the ordinary FGSM update itself provides an early warning. It computes an input gradient at the attack starting point and an endpoint loss gradient during parameter backpropagation. Retaining the endpoint input gradient exposes their cosine similarity without another model evaluation. We use "attack-free monitor" to emphasize that the monitor itself does not run any auxiliary validation attack. The base training still generates FGSM adversarial examples for parameter updates. When the cosine-drop score crosses a frozen threshold, the next mini-batch switches to two-step PGD and all later updates continue with PGD-2. A matched replay ablation tests whether restoring the affected epoch adds value beyond immediate action.

The contributions are:

- A gradient-consistency safety trigger calibrated on three CIFAR-10 development seeds and audited on seven disjoint held-out events. We also test unchanged-threshold architecture transfer and protocol transfer after one CIFAR-100-specific calibration.
- An event-level timing study that records warning, unprotected CO, and actual PGD-2 start times. Immediate switching acts before collapse in 5/5 matched events both with and without replay, compared with 3/5 for next-epoch escalation, isolating prompt action as the timing-critical part.
- Five-seed component controls and controlled FastAdv+/GradAlign reimplementations under one harness, with paired bootstrap intervals for small performance differences.
- A random-start stress test: the monitor stays silent on eight stable $8/255$ runs but detects 3/3 natural CO events at $16/255$, where immediate PGD-2 restores nontrivial robustness.

The study concerns empirical ℓ_∞ robustness. Certified defenses provide formal guarantees through different mechanisms and are complementary to the training-time monitor studied here.

II. Related Work

A. Adversarial Attacks and Training

FGSM follows the sign of the input loss gradient in one step [2]. PGD iterates gradient steps with projection onto a prescribed threat set and is central to the robust-optimization view of adversarial training [3]. TRADES separates natural error from a boundary-related robustness term and makes the clean–robust trade-off explicit [7]. Although these methods are effective, their multi-step inner loops increase training

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cost. Free adversarial training reuses gradients across repeated mini-batches [8], whereas fast training uses a single randomized FGSM step [4].

Robustness evaluation itself must be sufficiently strong. PGD results can be sensitive to iterations, restarts, and objectives. AutoAttack combines complementary parameter-free attacks to reduce evaluator bias and expose gradient masking [9]. In the controlled study below, we use progressively stronger full-test PGD variants and multiple restarts rather than treating FGSM accuracy as a robustness estimate.

B. Catastrophic Overfitting

Wong *et al.* identified CO as a failure mode of fast adversarial training [4]. Andriushchenko and Flammarion connected it to local nonlinearity and proposed GradAlign, which regularizes gradient alignment inside the perturbation set [6]. Li *et al.* showed that recovery from weak-attack overfitting is important and proposed FastAdv+ and FastAdvW, which use periodic PGD validation to decide when to introduce PGD updates [5]. Our response mechanism is related in spirit, but the detector differs operationally: it consumes the two input gradients already present in the ordinary FGSM parameter update and does not execute a validation attack. We also separate prediction from mitigation by freezing the detector before held-out evaluation.

C. Empirical and Certified Robustness

Empirical defenses are evaluated against known attacks but do not guarantee correctness for every allowed perturbation. Certified methods instead compute a provable lower bound, for example through convex outer approximations [10] or randomized smoothing under an ℓ_2 threat model [11]. This paper targets efficient empirical training in the ℓ_∞ setting. The distinction is important: the attack results reported below measure resistance to a strong test suite, not certified radii.

III. Method

A. Threat Model and Fast Update

Let f_θ be a classifier and ℓ cross-entropy loss. Adversarial training under an ℓ_∞ radius ϵ approximates

$$\min_{\theta} \mathbb{E}_{(x,y)} \left[\max_{\|\delta\|_\infty \le \epsilon} \ell(f_\theta(x+\delta), y) \right]. \tag{1}$$

FGSM begins at x^0, either x or a random point in the perturbation set, and constructs

$$g^0 = \nabla_{x^0} \ell(f_\theta(x^0), y), \quad x^{\mathrm{adv}} = \Pi\left(x^0 + \alpha \operatorname{sign}(g^0)\right), \tag{2}$$

where Π projects onto the valid image and threat set.

B. Attack-Free Gradient-Consistency Score

The endpoint backward pass produces $g^1 = \nabla_{x^{\mathrm{adv}}} \ell(f_\theta(x^{\mathrm{adv}}), y)$ while computing the parameter gradient. We retain it and calculate

$$c(x) = \frac{\langle g^0, g^1 \rangle}{\|g^0\|_2 \|g^1\|_2}. \tag{3}$$

FGSM relies on a local first-order approximation. A sharp decrease in the cosine between g^0 and g^1 means that the input gradient rotates along the FGSM direction, so the local linear approximation and the representativeness of the single-step inner maximizer are degrading. This interpretation agrees with prior links between local nonlinearity and CO [6]. We do not claim that the cosine drop is a necessary or sufficient condition for CO; it is an empirically validated proxy for the local-linearity failure associated with CO.

Batch averages are accumulated over non-overlapping windows of 20 optimizer updates to give c_t. The monitor adds reductions and trace state, but no forward or backward pass. In the implementation, g^0 is the gradient already needed to construct the FGSM example, while g^1 is retained from the normal parameter backward; the monitor calls no additional autograd-gradient operation. To remove the slow training trend, it uses

$$b_t = \operatorname{median}(c_{t-5}, \ldots, c_{t-1}), \qquad s_t = b_t - c_t, \tag{4}$$

after at least three history windows. A warning occurs when $s_t \ge \tau$. For each dataset calibration, we pool every finite development score, label a window positive only when a registered CO event is the next 20-update window, scan the unique observed scores, and choose the threshold maximizing Youden's J (true-positive rate minus false-positive rate). For CIFAR-10, seeds 17, 23, and 42 form the development set; this yields $\tau = 0.1663298875$, which is then frozen. CIFAR-100 uses the same fitting algorithm but a separate set of six dataset-specific development trajectories, as detailed in Section IV. Offline sensitivity analysis in the Appendix does not alter either fitted value or the main protocol.

C. Immediate Conditional Escalation

At the first warning, the primary response retains the completed FGSM update, switches the next mini-batch to PGD-2, and uses PGD-2 for all subsequent updates. Thus no additional FGSM update intervenes between warning and action, and no training state must be restored.

We compare this response with four matched alternatives. The replay variant keeps an in-memory epoch-start snapshot of model, optimizer, mixed-precision scaler, Python, NumPy, PyTorch CPU/CUDA, and loader-generator random states. At the warning it restores that state and replays the epoch using PGD-2; the snapshot is not serialized to disk. The remaining controls retain the current FGSM epoch and start PGD-2 at the next epoch, switch at a fixed early or late epoch, or use PGD-2 from epoch 1. These controls separate the value of prompt action from any effect of restoring the epoch.

D. Event and Warning/Evaluation Protocol

On fixed validation data, a CO event requires PGD-10 accuracy to fall at least 20 percentage points from its previous peak and reach at most 10%. We report whether a warning precedes the first event, whether it occurs within 60 updates, median lead, and alarm episodes in non-event runs. Validation PGD is used only for retrospective event labels; it never triggers the proposed monitor. Recall intervals are Wilson 95% intervals. Accuracy and time comparisons use identical seeds and paired bootstrap 95% intervals with 100,000 resamples.

IV. Experiments

A. Setup and Seed Roles

The primary setting is CIFAR-10 with PreActResNet-18 [12], [13], a deterministic 45,000/5,000 training/validation split, and $\epsilon = 8/255$. All methods use SGD with momentum 0.9, weight decay 5×10^{-4}, batch size 128, mixed precision, 30 epochs, and a triangular learning-rate schedule. The collapse-prone recipe uses zero-start FGSM, step $8/255$, and peak learning rate 0.3. The ordinary FGSM-RS control uses a uniform random start, step $10/255$, and peak learning rate 0.2. Recovery uses PGD-2 with step $4/255$.

We keep experimental roles explicit. Detector development uses CIFAR-10 seeds 17, 23, and 42. Held-out detector evaluation uses seeds 101, 202, 303, 404, 505, 606, and 707. Intervention ablations and baseline comparisons use the matched set 17, 23, 42, 101, and 202. Non-event controls are reported separately from event recall. Seeds 17, 23, and 42 are used for threshold calibration. They appear in intervention ablations but are not counted toward held-out detector-evaluation recall.

Transfer tests change one factor at a time. CIFAR-10/ResNet-18 uses the unchanged CIFAR-10 threshold on event seeds 303–707 and three non-event runs at seeds 17, 23, and 42. CIFAR-100 uses PreActResNet-18 and the same 20-update windows, five-window median, and three-window minimum history. Saved development traces at seeds 17, 23, and 42 set a one-time dataset-specific threshold of 0.6332649834303036 before evaluation of held-out event candidates 101, 202, and 303 and two random-start non-events. A separate CIFAR-10 threat-budget stress test retains random starts and uses $\epsilon = 16/255$, FGSM step $20/255$, and peak learning rate 0.2.

The CIFAR-100 calibration uses six development trajectories: one collapse-prone and one random-start run for each of seeds 17, 23, and 42. We apply the same window-level Youden-J algorithm used on CIFAR-10. Only one of the three collapse-prone development runs meets the registered CO definition, giving one positive next-window label among 2,877 eligible windows. The fitted threshold is 0.6332649834303036. The fitting rule is shared across datasets; the development data and resulting operating point are dataset-specific.

B. Held-Out Detection and Transfer

The primary held-out CIFAR-10 evaluation contains seven registered events spanning epochs 13–24, and none uses a threshold-development seed. The frozen monitor is active within 60 updates before all 7/7 events; the median lead of the earliest active warning inside that 60-update window is 40 updates. One event trajectory also contains an earlier alarm episode. The three ordinary FGSM-RS non-event runs contain no alarm episode. On the same event traces, a periodic PGD-10 probe every 100 updates detects 6/7 events with a median delay of 40 updates after the event and makes a median of 73 PGD calls before failure. The 100-update probe is a retrospective detector comparator applied to the saved collapse trajectories; it does not alter training. By contrast, the controlled FastAdv+ reimplementation probes a fixed 128-image validation batch every 20 training updates and uses the observed accuracy drop to trigger a transient PGD training block. The two frequencies therefore describe different protocols and computational budgets.

Table I separates observed events from all non-events. With the CIFAR-10 threshold unchanged, ResNet-18 yields 4/5 detected events but only 2/5 within 60 updates; this is partial unchanged-threshold architecture transfer, not complete cross-architecture generalization. On CIFAR-100, two of three planned collapse-prone trajectories meet the event definition and both are detected after the one-time dataset-specific calibration. The third collapse-prone trajectory does not meet the registered CO definition and is therefore counted with the two random-start controls: among these three non-events there is one alarm episode.

C. Randomized-FGSM Stress Test

We first test eight randomized-FGSM runs at $\epsilon = 8/255$: three standard runs at peak learning rate 0.2 and five runs at 0.3. All eight remain robust, and the detector produces zero alarm episodes. The monitor does not intervene in this tested standard $8/255$ configuration.

Prior work reports that random starts become less reliable at larger budgets [6]. At $\epsilon = 16/255$, all three random-start runs meet the CO definition and terminate at 0% PGD-10. The frozen CIFAR-10 detector warns 20, 40, and 60 updates before collapse. The intervention logs identify these three runs as the immediate epoch-replay variant, not the no-replay variant. Epoch replay begins PGD-2 before every event and finishes at 25.55%, 26.80%, and 27.11% PGD-10. Mean clean and PGD-10 accuracy are $71.05 \pm 0.43\%$ and $26.48 \pm 0.83\%$. This is a threat-budget stress test, not cross-dataset evidence.

D. Actionable Lead and Component Controls

Table II distinguishes statistical lead from usable lead. All three responses receive the identical first warning. Immediate switching begins PGD-2 on the next mini-batch, with no intervening FGSM update, and acts before all five matched collapses whether or not the epoch-start snapshot is restored. Waiting for the epoch boundary adds a median of 212 FGSM updates; two collapses occur before PGD-2 starts.

All component controls use seeds 17, 23, 42, 101, and 202. Immediate switching without replay differs from replay by $+0.22$ PGD-10 points (paired bootstrap 95% CI $[-0.22, +0.66]$) and $+0.06$ core minutes (CI $[-0.18, +0.30]$); the clean difference is -0.07 points (CI $[-0.16, 0.00]$). Neither comparison supports a replay advantage. Relative to waiting for the next epoch, it starts PGD-2 157.6 updates earlier on average (CI $[-252.0, -56.8]$), covers 5/5 rather than 3/5 events before CO, with a proportion difference of $+0.40$ (CI $[0.00, +0.80]$), and differs by $+0.33$ PGD-10 points (CI $[+0.08, +0.66]$). Its clean difference is -0.02 points (CI $[-0.14, +0.08]$), while core time increases by 0.59 minutes (CI $[+0.47, +0.71]$). The before-CO counts

TABLE I

DETECTOR AUDIT. "NEAR" DENOTES AN ACTIVE WARNING WITHIN 60 UPDATES. THE LAST COLUMN IS THE MEDIAN NEAR-WARNING LEAD IN UPDATES.

Dataset / budget	Architecture	Planned	Events	Detected	Near	Non-event runs / alarm episodes	Median near lead
CIFAR-10, $8/255$	PreActResNet-18	7	7	7/7	7/7	3 / 0	40
CIFAR-10, $8/255$	ResNet-18	5	5	4/5	2/5	3 / 0	80
CIFAR-100, $8/255$	PreActResNet-18	3	2	2/2	2/2	3 / 1	40
CIFAR-10, $16/255$ RS	PreActResNet-18	3	3	3/3	3/3	–	40

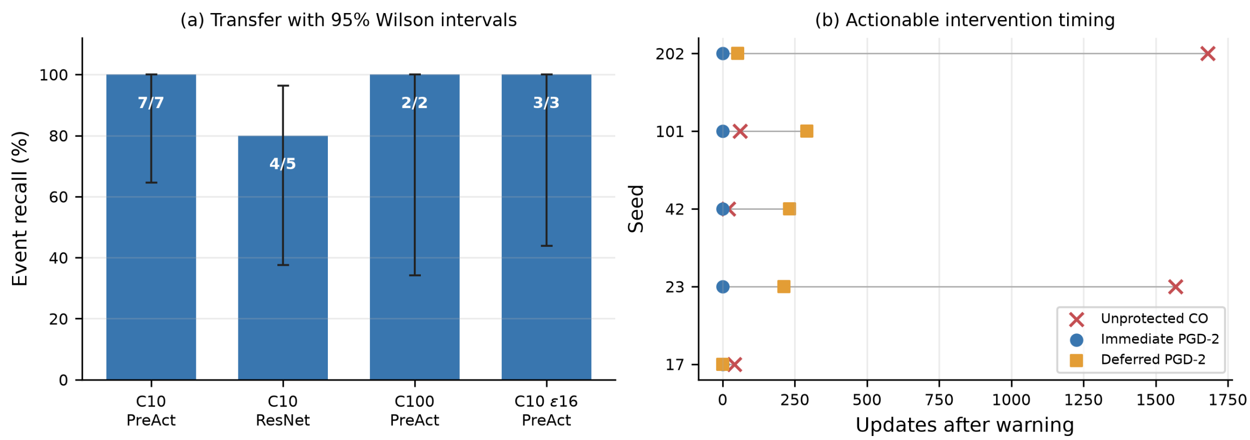

Fig. 1. Left: event recall across the primary setting, unchanged-threshold architecture transfer, dataset-calibrated transfer, and a threat-budget stress test, with Wilson 95% intervals. Right: both immediate variants start PGD-2 with zero intervening FGSM updates; next-epoch escalation starts after unprotected CO in two of five matched trajectories.

TABLE II

FIVE-SEED WARNING-TO-ACTION AUDIT (MEAN $\pm$ SAMPLE STD.). WARN→START COUNTS INTERVENING FGSM OPTIMIZER UPDATES.

Response	Before CO	Warn→start	Clean	PGD-10	Core min
Immediate, no replay	5/5	0 updates	83.06 ± 0.65	47.05 ± 0.74	18.74 ± 0.63
Immediate epoch replay	5/5	0 updates	83.14 ± 0.62	46.83 ± 0.55	18.68 ± 0.70
Next epoch	3/5	212 updates	83.09 ± 0.57	46.72 ± 1.05	18.14 ± 0.58

TABLE III

MATCHED FIVE-SEED CONTROLS (MEAN $\pm$ SAMPLE STD.)

Policy	Clean	PGD-10	Core min
Immediate, no replay	83.06 ± 0.65	47.05 ± 0.74	18.74 ± 0.63
Immediate replay	83.14 ± 0.62	46.83 ± 0.55	18.68 ± 0.70
Next epoch	83.09 ± 0.57	46.72 ± 1.05	18.14 ± 0.58
Fixed epoch 13	83.32 ± 0.70	46.41 ± 0.50	19.37 ± 0.21
Fixed epoch 21	82.35 ± 1.12	45.75 ± 0.94	17.85 ± 0.21
PGD-2 from epoch 1	83.57 ± 1.02	46.81 ± 0.90	21.60 ± 0.13

favor immediate switching, but five seeds are insufficient to establish a clear proportional advantage. The small endpoint difference is secondary to the intervention-timing result. Full PGD-2 differs from the next-epoch policy by $+0.09$ points (CI $[-0.27, +0.42]$) but adds 3.46 core minutes. The detector supplies the warning, PGD-2 supplies recovery, and immediate switching converts warning into pre-collapse action; rollback itself is not the active ingredient.

E. Controlled CO-Mitigation Baselines

FastAdv+ and GradAlign are controlled reimplementations rather than claims of using official released code. They share the split, architecture, five seeds, 30-epoch schedule, checkpoint rule, and PGD-10 evaluator. FastAdv+ uses one fixed 128-image validation batch, a PGD-10 probe every 20 optimizer updates with step $2/255$, a 0.10 drop from the best observed probe accuracy as its trigger, and a transient 20-update PGD-10 training block before returning to random-start FGSM. GradAlign uses $\lambda = 0.2$, a uniform perturbation in $[-\epsilon, \epsilon]$ clipped to the image domain, cosine alignment between clean and random-point input gradients, and a second-order graph only for the random branch. It otherwise uses batch size 128 and the shared SGD, momentum, weight decay, and triangular learning-rate schedule.

Table IV reports measured core parameter-update time. FastAdv+'s separately logged online PGD probes add 3.14 ± 0.00 minutes, giving 30.41 ± 3.53 measured minutes including those probes. Immediate replay's snapshot save-restore operations were not timed, so 18.68 minutes is not a complete end-to-end measurement. GradAlign achieves the highest mean PGD-10 accuracy among these 30-epoch methods; the proposed policy occupies a lower-robustness, lower-measured-core-cost point.

TABLE IV

UNIFIED FIVE-SEED CO-MITIGATION BASELINES

Method	Clean	PGD-10	Core min
Ours, next epoch	83.09 ± 0.57	46.72 ± 1.05	18.14 ± 0.58
Ours, immediate/no replay	83.06 ± 0.65	47.05 ± 0.74	18.74 ± 0.63
FastAdv+	81.96 ± 0.60	47.09 ± 0.71	27.27 ± 3.53
GradAlign	79.54 ± 0.86	48.31 ± 0.81	45.71 ± 0.11

TABLE V

FULL-TEST STRONG CIFAR-10 ACCURACY (%)

Method	Clean	PGD-10	PGD-50	PGD×5
Immediate, no replay (ours)	82.95	46.93	45.52	45.66
Epoch replay (ablation)	82.97	47.20	45.46	45.65
Next epoch	82.57	46.38	44.75	44.91
Fast FGSM-RS	82.48	47.77	46.24	46.36
PGD-10 training	82.47	51.76	50.77	50.84

F. Strong Test-Set Evaluation

Table V reports full-test clean and PGD evaluation on all 10,000 CIFAR-10 test images. Table VI gives a supplementary AutoAttack-standard evaluation on the first 1,000 examples in the unshuffled official test-set order. AA-1k uses the ℓ_∞ threat model at $\epsilon = 8/255$ and the standard attack suite comprising APGD-CE, APGD-T, FAB-T, and Square. The no-replay, replay, next-epoch, FastAdv+, and GradAlign checkpoints use seed 101; the FGSM-RS and PGD-10 training references use seed 17. In each run, *best_robust.pt* is selected by the highest recorded validation PGD-10 accuracy, before test evaluation. Reporting the seeds and fixed selection rule makes this a representative checkpoint comparison rather than a cross-seed aggregate. In particular, the no-replay seed-101 checkpoint was fixed before any result in Tables V–VI was observed; no alternative seed or checkpoint was tested for replacement. Table II's $47.05 \pm 0.74\%$ is instead the mean and sample standard deviation across five intervention seeds. The single-checkpoint and five-seed summaries answer different questions and need not match exactly.

For the primary no-replay method, PGD-50 and five-restart PGD-20 give 45.52% and 45.66%. Their agreement, and the analogous agreement for replay and the training references, does not reveal an obvious weak-iterative-attack failure. On the same fixed 1,000 images, complete AutoAttack-standard gives 42.30% for the primary no-replay method, 42.60% for replay, 41.30% for next-epoch escalation, 43.00% for ordinary FGSM-RS, 48.70% for PGD-10 training, 42.40% for FastAdv+, and 44.40% for GradAlign. The two controlled baselines use their seed-101 checkpoints selected by the same validation rule. AA-1k is a subset estimate, so it complements the full-test PGD results without ruling out gradient masking.

V. DISCUSSION, COST, AND LIMITATIONS

A. Causal and Temporal Interpretation

The experiments separate three causal roles. The detector supplies a warning, PGD-2 restores robustness, and immediate switching converts the warning into timely action so that

TABLE VI

AUTOATTACK-STANDARD ACCURACY ON THE FIXED FIRST 1,000 CIFAR-10 TEST IMAGES (%), ℓ_∞, $\epsilon = 8/255$

Method	AA-1k
Immediate, no replay (ours)	42.30
Epoch replay (ablation)	42.60
Next epoch	41.30
Fast FGSM-RS	43.00
PGD-10 training	48.70
FastAdv+	42.40
GradAlign	44.40

TABLE VII

EVENT-LEVEL TIMING IN OPTIMIZER UPDATES

Seed	Warn→CO	Immediate	Deferred	Deferred before CO
17	40	0	0	yes
23	1568	0	212	yes
42	20	0	232	no
101	60	0	292	no
202	1680	0	52	yes

PGD-2 can begin before CO. The no-replay control matches replay's 5/5 pre-collapse interventions and differs by only $+0.22$ PGD-10 points, with a paired interval crossing zero. Restoring the epoch snapshot is therefore not required for the observed timing benefit.

Table VII reports every matched timing event. High lead time is desirable for safety, but a warning far ahead of collapse does not imply precise prediction of collapse timing; the monitor acts as a safety trigger upon observing geometry degradation. In particular, the 1568- and 1680-update leads in seeds 23 and 202 should not be interpreted as calibrated time-to-CO forecasts. The saved-trace audit in the Appendix also checks whether the score remains above threshold after these early warnings.

B. Engineering Cost and Reproducibility

The normal FGSM update already computes the attack-start gradient and the endpoint parameter backward. The monitor retains the endpoint input gradient only until its cosine is reduced to a batch scalar, then stores scalar window history. It does not call a validation attack or an extra model evaluation.

The primary no-replay response does not require a snapshot. Its 18.74-minute measurement is still core training time, not complete end-to-end runtime. The replay control has a separate systems cost: its implementation holds one epoch-start copy of model, optimizer, and AMP-scaler state in device memory, plus Python, NumPy, PyTorch CPU/CUDA, and loader-generator RNG state. It does not write this replay snapshot to disk. Reported core training time does not include snapshot save-restore overhead. It does include abandoned FGSM parameter-update time and the replayed PGD-2 epoch. The experiment did not instrument snapshot-copy latency or peak memory, so the current measurements must not be read as complete end-to-end latency or memory results.

Every detector choice is fixed before its associated held-out evaluation: 20-update windows, five-window median history, three-window minimum history, the event definition, and the 60-update near horizon. Fixed-switch controls ignore detector callbacks. Retrospective PGD labels never trigger the proposed monitor. Seed-level warning, collapse, PGD-start, accuracy, and timing records remain in the saved experiment CSV files; paired intervals use 100,000 resamples.

C. Scope and Limitations

The architecture result is partial unchanged-threshold transfer: ResNet-18 detects 4/5 events but only 2/5 near events. CIFAR-100 is not zero-shot or plug-and-play; transfer requires six dataset-specific development trajectories to refit the detector operating point with the shared Youden-J algorithm. Its development set contains only one positive next-window label among 2,877 eligible windows, so the statistical stability of this calibration is limited. Event samples for ResNet-18, CIFAR-100, and the $16/255$ stress test remain small, as reflected by wide Wilson intervals. Transfer evidence is still concentrated in CIFAR datasets and the ResNet family; ImageNet and substantially different architectures remain outside the current scope.

The primary no-replay checkpoint is included in full-test PGD-50 and multi-restart PGD, plus complete AutoAttack-standard on a fixed 1,000-image subset, but not full-test AutoAttack. The subset is identical across compared checkpoints and includes APGD-CE, APGD-T, FAB-T, and Square. Agreement among PGD and this supplementary evaluation reduces, but does not eliminate, concern about gradient masking. The cosine score has empirical and mechanistic support but is neither a theoretical certificate nor a necessary or sufficient condition for CO. Finally, immediate replay introduces snapshot memory, storage-policy, and engineering overhead that has not been systematically characterized for larger models. The equally timely no-replay variant avoids this snapshot but has so far been evaluated only in the five-seed intervention ablation.

The controlled baselines identify a robustness-cost trade-off. GradAlign has higher mean PGD-10 robustness than the proposed policy. The policy uses lower measured core training time and provides an explicit event log, but the choice between them depends on whether the deployment prioritizes peak robustness, measured training cost, or auditable failure detection.

VI. Conclusion

We presented a gradient-consistency safety trigger for catastrophic overfitting and tested whether its warning arrives soon enough to change training. Seven held-out CIFAR-10/PreActResNet-18 events are detected within 60 updates. Transfer is partial with an unchanged threshold on ResNet-18 and requires dataset-specific development traces on CIFAR-100, although both datasets use the same Youden-J fitting algorithm. In a randomized-FGSM $16/255$ stress test, all three events are detected; the tested epoch-replay response raises terminal PGD-10 from zero to $26.48 \pm 0.83\%$.

For the pre-fixed seed-101 checkpoint of the primary no-replay response, full-test PGD-50 and five-restart PGD-20 accuracy are 45.52% and 45.66%, and AutoAttack-standard on the fixed first 1,000 test images gives 42.30%.

Across five seeds, the detector supplies the warning, PGD-2 restores robustness, and immediate switching determines whether recovery starts in time. Switching on the next mini-batch precedes CO in 5/5 cases with or without replay; rollback has no reliable final-accuracy advantage. In the controlled baselines, GradAlign obtains higher robustness and conditional escalation uses less measured core training time. The method is an empirical early-warning and conditional-computation mechanism, without a universal prevention claim, calibrated time-to-collapse output, or formal robustness guarantee.

Appendix

A. Immediate-Response Paired Intervals

Table VIII reports all requested paired bootstrap comparisons for the five-seed no-replay experiment. Accuracy differences are percentage points; "before CO" is a difference in proportions. Each interval uses 100,000 paired resamples of seeds 17, 23, 42, 101, and 202.

TABLE VIII
NO REPLAY MINUS THE MATCHED RESPONSE: MEAN DIFFERENCE [95% CI]

Metric	Epoch replay	Next epoch
Before-CO proportion	$0.00[0.00, 0.00]$	$+0.40[0.00, +0.80]$
Warn→start (updates)	$0.0[0.0, 0.0]$	$-157.6[-252.0, -56.8]$
Clean (points)	$-0.07[-0.16, 0.00]$	$-0.02[-0.14, +0.08]$
PGD-10 (points)	$+0.22[-0.22, +0.66]$	$+0.33[+0.08, +0.66]$
Core time (min)	$+0.06[-0.18, +0.30]$	$+0.59[+0.47, +0.71]$

B. Offline Sensitivity Analysis

Below we perform offline sensitivity analysis on saved held-out traces, without retraining or modifying main-paper frozen parameters. The main protocol remains $\tau = 0.1663298875$, a 60-update near horizon, and the original CO definition. The threshold scan is descriptive and is not used to replace the frozen value. It uses the seven held-out CIFAR-10 event trajectories and three separately reported randomized-FGSM non-event trajectories.

1) Threshold Sensitivity: Table IX recomputes warnings from saved cosine traces. Event recall, 60-update near recall, and non-event alarms are unchanged from 0.10 through 0.24. The median first-warning lead changes from 60 to 40 updates at thresholds of 0.20 and above. Event and false-alarm counts are stable around the frozen threshold; no post-hoc retuning is applied. Here "median first-warning lead" uses the first active warning anywhere before each event. Table I instead reports "median near lead": the earliest active warning restricted to the final 60 updates. At the frozen threshold these deliberately different summaries are 60 and 40 updates, respectively.

TABLE IX

OFFLINE THRESHOLD SENSITIVITY ON SAVED TRACES. LEAD IS THE FIRST-WARNING LEAD; TABLE I REPORTS NEAR LEAD.

τ	Recall	Near	Non-event alarms	Median first lead
0.10	7/7	7/7	0	60
0.12	7/7	7/7	0	60
0.14	7/7	7/7	0	60
0.16	7/7	7/7	0	60
0.1663298875 (frozen)	7/7	7/7	0	60
0.18	7/7	7/7	0	60
0.20	7/7	7/7	0	40
0.22	7/7	7/7	0	40
0.24	7/7	7/7	0	40

2) Warning-Horizon Sensitivity: At the frozen threshold, Table X varies only the reporting horizon. Six of seven events have an active warning within 20 updates, and all seven are covered by 40 updates. The registered 60-update horizon is retained in the main paper.

TABLE X

OFFLINE WARNING-HORIZON SENSITIVITY

Horizon (updates)	Near events	Near recall	Non-event alarms
20	6/7	85.7%	0
40	7/7	100%	0
60 (frozen)	7/7	100%	0
80	7/7	100%	0
100	7/7	100%	0

3) CO-Definition Sensitivity: The original event requires a drop of at least 0.20 from the previous PGD-10 peak and accuracy at most 0.10. Table XI changes these labeling constants only after training. All seven trajectories remain events and are detected before the event under every variant. Near recall changes because stricter labels place the registered event later or earlier relative to saved warning windows; the main definition is unchanged.

TABLE XI

OFFLINE CO EVENT-DEFINITION SENSITIVITY

Definition	Drop	Ceiling	Events	Detected	Near
Relaxed	0.15	0.15	7	7/7	7/7
Original	0.20	0.10	7	7/7	7/7
Higher ceiling	0.20	0.15	7	7/7	7/7
Strict	0.25	0.05	7	7/7	5/7

4) Long-Lead Signal Audit: Table XII checks the two long-lead intervention traces. The score is not continuously above threshold: only four post-warning windows are active in each case. The first crossing is treated as a safety trigger after observed geometry degradation, not as a calibrated estimate of time to collapse.

TABLE XII

SIGNAL PERSISTENCE AFTER LONG-LEAD WARNINGS

Seed	Lead	Windows	Active	Longest active run
23	1568	80	4 (5.0%)	3
202	1680	86	4 (4.7%)	2

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